LAB 04

1. List the user commands and redirect the output to /tmp/commands.list

```
[mina_shafik@localhost ~]$ history > /tmp/commands.list
[mina_shafik@localhost ~]$ cat /tmp/commands.list
    1  groupadd -g 30000 pgroup
    2  groupadd -g 30000 pgroup
    3  root
    4  su root
    5  clear
    6  groupadd -g 30000 pgroup
    7  groups mina_shafik
    8  clear
    9  groupdel pgroup
   10  sudo groupdel pgroup
   11  sudo groupdel badgroup
   12  groups mina_shafik
   13  getent group pgroup
   14  getent group badgroup
   15  groups mina_shafik
   16  clear
```

2. Count the number of user commands

```
[mina_shafik@localhost ~]$ wc -l /tmp/commands.list
243 /tmp/commands.list
[mina_shafik@localhost ~]$
```

3. Get all the users names whose first character in their login is 'g'.

```
[mina_shafik@localhost ~]$ grep '^g' /etc/passwd
games:x:12:100:games:/usr/games:/sbin/nologin
[mina_shafik@localhost ~]$
```

4. Get the logins name and full names (comment) of logins starts with "g".

```
[mina_shafik@localhost ~]$ grep '^g' /etc/passwd | cut -d: -f1,5
games:games
[mina_shafik@localhost ~]$
```

5. Save the output of the last command sorted by their full names in a file.

```
[mina_shafik@localhost ~]$ grep '^m' /etc/passwd | cut -d: -f1,5 | sort -t: -k2 > /tmp/users.
list
[mina_shafik@localhost ~]$ cat /tmp/users.list
mina:
mail:mail
mina_shafik:mina shafik iti
[mina_shafik@localhost ~]$
```

6. Display the number of users who is logged now to the system.

```
[mina_shafik@localhost ~]$ who | wc -l
2
[mina_shafik@localhost ~]$ who
root     tty1         2026-03-14 17:34
root     pts/0        2026-03-14 18:51 (192.168.56.1)
[mina_shafik@localhost ~]$
```

7. Display lines 7 to line 10 of /etc/passwd file

```
[mina_shafik@localhost ~]$ sed -n '7,10p' /etc/passwd
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
[mina_shafik@localhost ~]$
```

8. What happens if you execute:

a. cat filename1 | cat filename2

```
[mina_shafik@localhost ~]$ cat filename1 | cat filename2
cat: cat: filename2: No such file or directory
filename1: No such file or directory
[mina_shafik@localhost ~]$
```

b. ls | rm

```
[mina_shafik@localhost ~]$ ls | rm
rm: missing operand
Try 'rm --help' for more information.
[mina_shafik@localhost ~]$
```

c. ls /etc/passwd | wc –l

```
[mina_shafik@localhost ~]$ ls /etc/passwd | wc -l
1
[mina_shafik@localhost ~]$
```

LAB 04 – Processes

1. Issue the command sleep 100.

```
[root@localhost ~]# sleep 100
^Z
[1]+  Stopped                 sleep 100
[root@localhost ~]#
```

2. Stop the last command.

```
[root@localhost ~]# sleep 100
^Z
[1]+  Stopped                 sleep 100
[root@localhost ~]#
```

3. Resume the last command in the background.

```
[root@localhost ~]# bg
[1]+ sleep 100 &
```

4. Issue the jobs command and see its output.

```
[root@localhost ~]# jobs
[1]+  Done                    sleep 100
[root@localhost ~]#
```

5. Send sleep command to foreground and send it again to the background.

```
[root@localhost ~]# sleep 100
^Z
[1]+  Stopped                 sleep 100
[root@localhost ~]# fg %1
sleep 100
```

6. Kill the sleep command.

```
[root@localhost ~]# kill %1

[1]+  Stopped                 sleep 100
[root@localhost ~]# pkill sleep
[1]+  Terminated              sleep 100
[root@localhost ~]#
```

7. Display your processes only.

```
[root@localhost ~]# ps
    PID TTY          TIME CMD
   1340 pts/0    00:00:00 bash
  15989 pts/0    00:00:00 ps
[root@localhost ~]#
```

8. Display all processes except yours.

```
[root@localhost ~]# ps
    PID TTY          TIME CMD
   1340 pts/0    00:00:00 bash
  15989 pts/0    00:00:00 ps
[root@localhost ~]# ps -u
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root        1312  0.0  0.2   7440  4216 tty1     Ss+  14:42   0:00 -bash
root        1340  0.0  0.2   7440  4352 pts/0    Ss   14:42   0:00 -bash
root       16170  0.0  0.1  10140  3620 pts/0    R+   15:25   0:00 ps -u
```

9. Use the pgrep command to list your processes only.

```
[root@localhost ~]# pgrep -u $USER -l
1 systemd
2 kthreadd
3 rcu_gp
4 rcu_par_gp
5 slub_flushwq
6 netns
8 kworker/0:0H-events_highpri
10 kworker/0:1H-events_highpri
```

10.Kill your processes only.

```
[root@localhost ~]# sleep 100
^Z
[1]+  Stopped                 sleep 100
[root@localhost ~]# pkill -u $USER sleep
[root@localhost ~]#
```

LAB 04: Exploring the Logging System

1. Display the last 20 lines of the system log file /var/log/messages.

```
[root@localhost ~]# tail -20 /var/log/messages
Mar 15 14:56:56 localhost systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Mar 15 14:56:56 localhost systemd[1]: Finished Cleanup of Temporary Directories.
Mar 15 14:56:56 localhost systemd[1]: run-credentials-systemd\x2dtmpfiles\x2dclean.service.mount: Deactivated successfully.
Mar 15 14:57:20 localhost NetworkManager[692]: <info>  [1773579440.4170] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
Mar 15 15:02:20 localhost NetworkManager[692]: <info>  [1773579740.4099] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
Mar 15 15:07:20 localhost NetworkManager[692]: <info>  [1773580040.4175] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
Mar 15 15:12:20 localhost NetworkManager[692]: <info>  [1773580340.4244] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
Mar 15 15:17:20 localhost NetworkManager[692]: <info>  [1773580640.4217] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
Mar 15 15:18:46 localhost systemd[1]: Starting dnf makecache...
Mar 15 15:18:53 localhost dnf[13726]: Rocky Linux 9 - BaseOS                        2.8 kB/s | 4.3 kB     00:01
Mar 15 15:18:54 localhost dnf[13726]: Rocky Linux 9 - AppStream                     3.1 kB/s | 4.8 kB     00:01
Mar 15 15:18:57 localhost dnf[13726]: Rocky Linux 9 - Extras                        1.8 kB/s | 3.1 kB     00:01
Mar 15 15:18:57 localhost dnf[13726]: Metadata cache created.
Mar 15 15:18:57 localhost systemd[1]: dnf-makecache.service: Deactivated successfully.
Mar 15 15:18:57 localhost systemd[1]: Finished dnf makecache.
Mar 15 15:18:57 localhost systemd[1]: dnf-makecache.service: Consumed 2.355s CPU time.
Mar 15 15:22:20 localhost NetworkManager[692]: <info>  [1773580940.4348] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
Mar 15 15:27:20 localhost NetworkManager[692]: <info>  [1773581240.4350] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
Mar 15 15:32:20 localhost NetworkManager[692]: <info>  [1773581540.4379] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
Mar 15 15:37:20 localhost NetworkManager[692]: <info>  [1773581840.6292] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
[root@localhost ~]#
```

It contains all details about the log and who logged what, when and how

2. Display last 10 lines of /var/log/secure , explain what this file contains.

```
[root@localhost ~]# su - mina_shafik
Last login: Sat Mar 14 18:52:15 EET 2026 on pts/0
[mina_shafik@localhost ~]$ sudo tail -n 10 /var/log/secure
[sudo] password for mina_shafik:
Mar 15 14:42:16 localhost sshd[706]: Server listening on 0.0.0.0 port 22.
Mar 15 14:42:16 localhost sshd[706]: Server listening on :: port 22.
Mar 15 14:42:29 localhost systemd[1303]: pam_unix(systemd-user:session): session opened for user root(uid=0) by (uid=0)
Mar 15 14:42:29 localhost login[710]: pam_unix(login:session): session opened for user root(uid=0) by LOGIN(uid=0)
Mar 15 14:42:29 localhost login[710]: ROOT LOGIN ON tty1
Mar 15 14:42:56 localhost sshd[1333]: Accepted password for root from 192.168.56.1 port 59379 ssh2
Mar 15 14:42:56 localhost sshd[1333]: pam_unix(sshd:session): session opened for user root(uid=0) by (uid=0)
Mar 15 14:42:56 localhost sshd[1337]: Accepted password for root from 192.168.56.1 port 59383 ssh2
Mar 15 14:42:56 localhost sshd[1337]: pam_unix(sshd:session): session opened for user root(uid=0) by (uid=0)
Mar 15 15:59:11 localhost su[27679]: pam_unix(su-l:session): session opened for user mina_shafik(uid=1051) by root(uid=0)
[mina_shafik@localhost ~]$
```

SSH login attempts (successful and failed)
User authentication events
sudo command usage
su (switch user) attempts
PAM authentication messages
Security warnings

3. Use the tail -f command to monitor /var/log/messages in real time.
Open another terminal and log in/out, then observe the output.

```
[mina_shafik@localhost ~]$ sudo tail -f /var/log/messages
Mar 15 16:05:46 localhost systemd[29807]: Finished Exit the Session.
Mar 15 16:05:46 localhost systemd[29807]: Reached target Exit the Session.
Mar 15 16:05:46 localhost systemd[1]: user@1051.service: Deactivated successfully.
Mar 15 16:05:46 localhost systemd[1]: Stopped User Manager for UID 1051.
Mar 15 16:05:46 localhost systemd[1]: Stopping User Runtime Directory /run/user/1051 ...
Mar 15 16:05:46 localhost systemd[1]: run-user-1051.mount: Deactivated successfully.
Mar 15 16:05:46 localhost systemd[1]: user-runtime-dir@1051.service: Deactivated successfully.
Mar 15 16:05:46 localhost systemd[1]: Stopped User Runtime Directory /run/user/1051.
Mar 15 16:05:46 localhost systemd[1]: Removed slice User Slice of UID 1051.
Mar 15 16:05:48 localhost systemd[1]: systemd-hostnamed.service: Deactivated successfully.
^Z
[3]+  Stopped                 sudo tail -f /var/log/messages
[mina_shafik@localhost ~]$ su
Password:
[root@localhost mina_shafik]# sudo tail -f /var/log/messages
Mar 15 16:05:46 localhost systemd[1]: user@1051.service: Deactivated successfully.
Mar 15 16:05:46 localhost systemd[1]: Stopped User Manager for UID 1051.
Mar 15 16:05:46 localhost systemd[1]: Stopping User Runtime Directory /run/user/1051 ...
Mar 15 16:05:46 localhost systemd[1]: run-user-1051.mount: Deactivated successfully.
Mar 15 16:05:46 localhost systemd[1]: user-runtime-dir@1051.service: Deactivated successfully.
Mar 15 16:05:46 localhost systemd[1]: Stopped User Runtime Directory /run/user/1051.
Mar 15 16:05:46 localhost systemd[1]: Removed slice User Slice of UID 1051.
Mar 15 16:05:48 localhost systemd[1]: systemd-hostnamed.service: Deactivated successfully.
Mar 15 16:07:09 localhost su[30472]: (to root) mina_shafik on pts/0
Mar 15 16:07:20 localhost NetworkManager[692]: <info>  [1773583640.4492] dhcp4 (enp0s8): state changed new lease, address=192.168.56.101
```

It’s about the system messages that if there are any changes like switching the user in out case

4. What is the difference between /var/log/messages and /var/log/dmesg?

`/var/log/messages`: General system log, System events, service messages, warnings, errors, network activity, and daemon logs

`/var/log/dmesg`: Kernel boot log,Messages generated by the **Linux kernel**, mainly during system startup and hardware detection

Using journalctl

5. Display all journal log entries using journalctl.

```
[mina_shafik@localhost ~]$ sudo journalctl
[sudo] password for mina_shafik:
Mar 15 14:41:54 localhost kernel: Linux version 5.14.0-284.11.1.el9_2.x86_64 (mockbuild@iad1-prod-build001.bld.equ.rockylinux.org) (gcc (GCC) 1
Mar 15 14:41:54 localhost kernel: The list of certified hardware and cloud instances for Enterprise Linux 9 can be viewed at the Red Hat Ecosys
Mar 15 14:41:54 localhost kernel: Command line: BOOT_IMAGE=(hd0,msdos1)/vmlinuz-5.14.0-284.11.1.el9_2.x86_64 root=/dev/mapper/rl-root ro crashk
Mar 15 14:41:54 localhost kernel: x86/fpu: x87 FPU will use FXSAVE
Mar 15 14:41:54 localhost kernel: signal: max sigframe size: 1440
Mar 15 14:41:54 localhost kernel: BIOS-provided physical RAM map:
Mar 15 14:41:54 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Mar 15 14:41:54 localhost kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Mar 15 14:41:54 localhost kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Mar 15 14:41:54 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x000000007ffeffff] usable
Mar 15 14:41:54 localhost kernel: BIOS-e820: [mem 0x000000007fff0000-0x000000007fffffff] ACPI data
Mar 15 14:41:54 localhost kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
Mar 15 14:41:54 localhost kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Mar 15 14:41:54 localhost kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
```

6. Display only kernel messages using journalctl.

```
[mina_shafik@localhost ~]$ journalctl -k -f
Mar 15 14:42:08 localhost kernel: fbcon: svgadrmfb (fb0) is primary device
Mar 15 14:42:08 localhost kernel: Console: switching to colour frame buffer device 100x37
Mar 15 14:42:08 localhost kernel: [drm] Initialized vmwgfx 2.20.0 20211206 for 0000:00:02.0 on minor 0
Mar 15 14:42:08 localhost kernel: snd_intel8x0 0000:00:05.0: allow list rate for 1028:0177 is 48000
Mar 15 14:42:08 localhost kernel: XFS (sda1): Starting recovery (logdev: internal)
Mar 15 14:42:08 localhost kernel: XFS (sda1): Ending recovery (logdev: internal)
Mar 15 14:42:16 localhost.localdomain kernel: e1000: enp0s3 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
Mar 15 14:42:16 localhost.localdomain kernel: IPv6: ADDRCONF(NETDEV_CHANGE): enp0s3: link becomes ready
Mar 15 14:42:16 localhost.localdomain kernel: e1000: enp0s8 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
Mar 15 14:42:17 localhost.localdomain kernel: IPv6: ADDRCONF(NETDEV_CHANGE): enp0s8: link becomes ready
```

7. Display logs from the current boot only.

```
[mina_shafik@localhost ~]$ journalctl -b -f
Mar 15 16:17:20 localhost.localdomain NetworkManager[692]: <info>  [1773584240.4468] dhcp4 (enp0s8): state changed new lease, address=192.168.56
.101
Mar 15 16:19:05 localhost.localdomain su[34572]: (to mina_shafik) root on pts/0
Mar 15 16:19:05 localhost.localdomain su[34572]: pam_unix(su-l:session): session opened for user mina_shafik(uid=1051) by root(uid=0)
Mar 15 16:19:05 localhost.localdomain systemd[1]: Starting Hostname Service...
Mar 15 16:19:05 localhost.localdomain systemd[1]: Started Hostname Service.
Mar 15 16:19:35 localhost.localdomain systemd[1]: systemd-hostnamed.service: Deactivated successfully.
Mar 15 16:21:42 localhost.localdomain sudo[35483]: mina_shafik : TTY=pts/0 ; PWD=/home/mina_shafik ; USER=root ; COMMAND=/bin/journalctl
Mar 15 16:21:42 localhost.localdomain sudo[35483]: pam_unix(sudo:session): session opened for user root(uid=0) by root(uid=1051)
Mar 15 16:22:20 localhost.localdomain NetworkManager[692]: <info>  [1773584540.4337] dhcp4 (enp0s8): state changed new lease, address=192.168.56
.101
Mar 15 16:27:20 localhost.localdomain NetworkManager[692]: <info>  [1773584840.4411] dhcp4 (enp0s8): state changed new lease, address=192.168.56
.101
```

8. Display logs from the previous boot.

```
[mina_shafik@localhost ~]$ journalctl --list-boots
IDX BOOT_ID                          FIRST_ENTRY                  LAST_ENTRY
  0 e51703e18d9d48e1803cff95843612a6 Sun 2026-03-15 14:41:54 EET Sun 2026-03-15 16:27:20 EET
[mina_shafik@localhost ~]$
```

9. Filter journal logs to show only error and critical priority messages.

```
[mina_shafik@localhost ~]$ journalctl -p err..crit
Mar 15 14:41:54 localhost systemd[1]: Invalid DMI field header.
Mar 15 14:41:55 localhost kernel: Warning: Unmaintained hardware is detected:  e1000:100E:8086 @ 0000:00:03.0
Mar 15 14:41:56 localhost kernel: Warning: Unmaintained hardware is detected:  e1000:100E:8086 @ 0000:00:08.0
Mar 15 14:42:05 localhost systemd[1]: Invalid DMI field header.
Mar 15 14:42:08 localhost kernel: [drm:vmw_host_printf [vmwgfx]] *ERROR* Failed to send host log message.
[mina_shafik@localhost ~]$
```

10.Display logs generated in the last 30 minutes.

```
[mina_shafik@localhost ~]$ journalctl --since "30 minutes ago"
Mar 15 16:02:20 localhost.localdomain NetworkManager[692]: <info>  [1773583340.4469] dhcp4 (enp0s8): state changed new lease, address=192.168.5
Mar 15 16:04:35 localhost.localdomain sshd[29482]: Connection closed by ::1 port 41854 [preauth]
Mar 15 16:05:14 localhost.localdomain systemd[1]: Starting SSSD Kerberos Cache Manager...
Mar 15 16:05:15 localhost.localdomain systemd[1]: Started SSSD Kerberos Cache Manager.
Mar 15 16:05:15 localhost.localdomain sssd_kcm[29785]: Starting up
Mar 15 16:05:18 localhost.localdomain sshd[29758]: Accepted password for mina_shafik from ::1 port 36272 ssh2
Mar 15 16:05:18 localhost.localdomain systemd[1]: Created slice User Slice of UID 1051.
Mar 15 16:05:18 localhost.localdomain systemd[1]: Starting User Runtime Directory /run/user/1051 ...
Mar 15 16:05:18 localhost.localdomain systemd-logind[673]: New session 5 of user mina_shafik.
Mar 15 16:05:18 localhost.localdomain systemd[1]: Finished User Runtime Directory /run/user/1051.
Mar 15 16:05:18 localhost.localdomain systemd[1]: Starting User Manager for UID 1051 ...
Mar 15 16:05:18 localhost.localdomain systemd[29807]: pam_unix(systemd-user:session): session opened for user mina_shafik(uid=1051) by (uid=0)
Mar 15 16:05:18 localhost.localdomain systemd[29807]: Queued start job for default target Main User Target.
Mar 15 16:05:18 localhost.localdomain systemd[29807]: Created slice User Application Slice.
Mar 15 16:05:18 localhost.localdomain systemd[29807]: Started Mark boot as successful after the user session has run 2 minutes.
```

11.Display logs for a specific service (e.g., sshd):

```
[mina_shafik@localhost ~]$ journalctl -u sshd
Mar 15 14:42:16 localhost systemd[1]: Starting OpenSSH server daemon...
Mar 15 14:42:16 localhost sshd[706]: Server listening on 0.0.0.0 port 22.
Mar 15 14:42:16 localhost systemd[1]: Started OpenSSH server daemon.
Mar 15 14:42:16 localhost sshd[706]: Server listening on :: port 22.
Mar 15 14:42:56 localhost.localdomain sshd[1333]: Accepted password for root from 192.168.56.1 port 59379 ssh2
Mar 15 14:42:56 localhost.localdomain sshd[1333]: pam_unix(sshd:session): session opened for user root(uid=0) by (uid=0)
Mar 15 14:42:56 localhost.localdomain sshd[1337]: Accepted password for root from 192.168.56.1 port 59383 ssh2
Mar 15 14:42:56 localhost.localdomain sshd[1337]: pam_unix(sshd:session): session opened for user root(uid=0) by (uid=0)
Mar 15 16:04:35 localhost.localdomain sshd[29482]: Connection closed by ::1 port 41854 [preauth]
Mar 15 16:05:18 localhost.localdomain sshd[29758]: Accepted password for mina_shafik from ::1 port 36272 ssh2
Mar 15 16:05:18 localhost.localdomain sshd[29758]: pam_unix(sshd:session): session opened for user mina_shafik(uid=1051) by (uid=0)
[mina_shafik@localhost ~]$
```